Route G-R12: Local Correlated Preparation and the Energy Cost of Its Mediator

An added controlled interaction; unchanged background contact

9 October 2026

Abstract

The G-R9b density-contact obstruction does not follow from translation symmetry alone. A local occupation-dependent hopping realizes the desired dark-clock preparation block without inserting a momentum projector. Keeping the old contact, one fixed pulse gives unconditional fidelities above 0.997 in the inherited fixtures. A finite, 21-state mediator model then exposes a leading Stark term and high-energy switching residue. This is a controlled preparation construction, not autonomous event or time generation; its drive phase, pulse and compensation are explicit.

1 A local interaction outside the density-only family

Use two distinguishable three-node rings, with $S|j\rangle = |j+1 \bmod 3\rangle$, and retain the full G-R10 system Hamiltonian

$$L_\theta = 2I - e^{\mathrm{i}\theta/3}S - e^{-\mathrm{i}\theta/3}S^\dagger, \tag{1}$$

$$H_\gamma = H_0 + \gamma P_0 \otimes P_0, \quad H_0 = L_{\pi/2} \otimes I + I \otimes L_{\pi/2}. \tag{2}$$

All nine product states remain present. In the site basis define

$$D = \sum_{j=0}^{2} P_j \otimes P_j, \quad K = L_0 \otimes I + I \otimes L_0, \qquad C = \frac{\mathrm{i}}{\sqrt{2}}[K, D]. \tag{3}$$

D detects coincidence, not an external density or Chen's E_d. For configurations $x = (j, l)$ and $y = (j', l')$, put $d_x = \delta_{jl}$. Then

$$C_{xy} = \frac{\mathrm{i}}{\sqrt{2}} K_{xy}(d_y - d_x). \tag{4}$$

Thus C contains only single-ring nearest-neighbour hops which change coincidence. There are 12 undirected edges (24 directed entries), not an arbitrary projector onto selected Fourier modes. Graph locality is not a derivation of physical space. $C = C^\dagger$ because $[K, D]$ is anti-Hermitian. Both K and D commute with joint translation $\mathsf{T} = S \otimes S$ and the swap X, so C does too.

K is a *new zero-flux hopping quadrature*, not the existing H_0. It is not generated merely by retiming the original contact. We add gC during preparation and account for its switching separately.

2 Exact preparation block and the precise symmetry correction

Let $F_{jk} = e^{2\pi \mathrm{i} jk/3}/\sqrt{3}$ and use the fixed Fourier basis. The joint-translation sector $a + b = 2 \bmod 3$ is exactly $\mathcal{T} = \mathrm{span}(|02\rangle, |11\rangle, |20\rangle)$. Writing Π for its projector, direct Fourier transformation gives

$$K_{\mathcal{T}} = \mathrm{diag}(3, 6, 3), \quad D_{\mathcal{T}} = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad C_{\mathcal{T}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & -\mathrm{i} & 0 \\ \mathrm{i} & 0 & \mathrm{i} \\ 0 & -\mathrm{i} & 0 \end{pmatrix}. \tag{5}$$

No coupling leaves this sector under C. Set

$$|M\rangle = |11\rangle, \quad |A\rangle = \frac{|02\rangle + |20\rangle}{\sqrt{2}}, \quad |D_+\rangle = \frac{1}{\sqrt{3}}|A\rangle - \sqrt{\frac{2}{3}}|M\rangle. \tag{6}$$

Then $C_{\mathcal{T}} = \mathrm{i}(|M\rangle\langle A| - |A\rangle\langle M|)$ and $H_{0,\mathcal{T}} = 4I$. Hence, for $\gamma = 0$ and up to an overall phase,

$$e^{-\mathrm{i}s(H_0 + gC)}|M\rangle = \cos(gs)|M\rangle - \sin(gs)|A\rangle, \quad \beta = \arcsin(1/\sqrt{3}), \quad gs_* = \beta, \tag{7}$$

which equals $-|D_+\rangle$ at s_*. For any area $a = gs$,

$$F_0(a) = \cos^2(a - \beta), \qquad F_0(-\beta) = 1/9. \tag{8}$$

The coupling sign is therefore physical with the original state and readout held fixed. At $a = \beta + \pi/2$ the target fidelity vanishes: continued control is not storage.

What G-R9b does and does not rule out. Its $h(j-l)$ density contacts give circulant blocks and conserve their common bright component. Equation (5) is not circulant, even though it commutes with T and X. Joint translation acts as a scalar throughout $\mathcal{T}$, so it cannot force a particular matrix within that sector. The 2/3 obstruction therefore applies to the specified contact family, not to every translation-invariant interaction. The old rank-one 4/9 bound remains valid and unchanged. Also $\Pi[H_0, D]\Pi = 0$: replacing the new K by the old resonant kinetic term would remove precisely the desired block.

An exact occupation-dependent Peierls identity further characterizes the new operator. For $K_\chi = e^{\mathrm{i}\chi D} K e^{-\mathrm{i}\chi D}$,

$$C = -\frac{K_\chi - K_{-\chi}}{2\sqrt{2}\sin\chi}, \qquad \sin\chi \neq 0. \tag{9}$$

It follows entry by entry since $d_x - d_y \in \{0, \pm 1\}$. Adding this difference while retaining H_0 is a new interaction. Conjugating every state, operator and probe together would instead be only a basis change.

3 Keep the original contact and count the controlled pulse

The actual direct-pulse test uses $H_{\mathrm{on}} = H_\gamma + gC$, not the isolated two-state block. Inherited parameters are $g = 0.1$ and $\alpha = 0.25$ per assumed update. Before evaluating any γ, fix

$$n = \mathrm{round}\,\frac{\beta}{g\alpha} = 25, \qquad s_f = n\alpha = 6.25 \quad (s_* = 6.15479709\ldots). \tag{10}$$

There is no pulse optimization for individual fixtures. s remains the declared evolution parameter, not physical seconds. Exact 9×9 exponentials give the following *unconditional* results:

γ	F_D	probability outside $\mathcal{T}$	net work on system
0	0.999909367	0	0
0.09	0.997140947	$1.78685\,10^{-4}$	−0.00992608
0.03	0.999600916	$1.97636\,10^{-5}$	−0.00333098
0.01	0.999875084	$2.19518\,10^{-6}$	−0.00111089

The original vertex contact breaks the sector closure, so leakage is retained rather than projected away. During the hold, the expectation of H_{on} is constant. Sudden switching gives the exact ledger

$$W_{\mathrm{on}} = g\langle M|C|M\rangle = 0, \quad W_{\mathrm{off}} = -g\langle\psi_f|C|\psi_f\rangle, \tag{11}$$

$$\langle H_\gamma\rangle_f - \langle H_\gamma\rangle_i = W_{\mathrm{on}} + W_{\mathrm{off}}, \qquad \langle H_\gamma\rangle_i = 4 + \gamma/9. \tag{12}$$

Negative work means energy is delivered to the controller. It does not mean that the pulse apparatus, switching precision or resetting is free. The Hamiltonian before and after the pulse is the same H_γ.

Why this does not contradict the old energy bound. For the inherited positive γ fixtures, the $E = 4$ eigenspace is $P_D = |D_+\rangle\langle D_+| + |D_-\rangle\langle D_-|$. Any system-only unitary commuting with H_γ preserves its weight, so

$$[U, H_\gamma] = 0 \quad \Longrightarrow \quad |\langle D_+|U|M\rangle|^2 \le \langle M|P_D|M\rangle = 2/3. \tag{13}$$

Our pulse is not such a unitary; the full work ledger is essential. An autonomous energy receiver would have to match the full energy distribution, not merely its mean.

Connection to the existing record. Swap symmetry leaves the antisymmetric D_- amplitude zero. Thus the unchanged G-R11 reader now succeeds with probability exactly F_D, not 2/3. Conditional on success its unread output is still $(57/82)|D_+\rangle\langle D_+| + (25/82)|D_-\rangle\langle D_-|$. The measurement disturbance has not disappeared. After pulse removal, F_D stays constant because D_+ is an eigenstate of H_γ; this does not make the entire imperfect output state stationary.

4 A finite local mediator, including its leading Stark term

To check more than a prescribed effective operator, introduce one auxiliary configuration state for each of the 12 undirected edges in Eq. (4). These are explicit additional degrees of freedom, initially empty, with common energy $4+\Delta$. For edge $e=(x,y)$, $x<y$, choose

$$B_{ex}=\Omega, \quad B_{ey}=-\frac{g\Delta C_{xy}}{\Omega}, \quad \Omega=\sqrt{\frac{g\Delta}{\sqrt{2}}}, \qquad B_{ez}=0 \ (z\neq x,y). \tag{14}$$

Both nonzero amplitudes have modulus Ω. Their relative phase is supplied; the construction does not derive its sign from symmetry. The full 21×21 hold Hamiltonian is

$$\mathcal{H}_{\mathrm{on}}=\begin{pmatrix} H_\gamma+V_c & B^\dagger \\ B & (4+\Delta)I_{12}\end{pmatrix}, \quad \mathcal{H}_{\mathrm{off}}=H_\gamma\oplus(4+\Delta)I_{12}. \tag{15}$$

Every auxiliary exit is retained in the exact propagation.

Eliminating auxiliaries from the eigenvalue equation gives the exact energy-dependent Schur operator

$$H_{\mathrm{low}}(E)=H_\gamma+V_c-\frac{B^\dagger B}{\Delta-(E-4)}, \tag{16}$$

$$H_{\mathrm{low}}(E)=H_\gamma+V_c-\frac{B^\dagger B}{\Delta}+R(E), \quad \|R(E)\|\leq\frac{\|B^\dagger B\||E-4|}{\Delta(\Delta-|E-4|)} \quad (|E-4|<\Delta). \tag{17}$$

This controlled low-energy expansion alone does not certify a suddenly switched initial state. The diagonal terms already matter at leading order. A coincident configuration has four mediator edges, a noncoincident one two. Consequently,

$$-\frac{B^\dagger B}{\Delta}=gC-\sqrt{2}g(I+D). \tag{18}$$

The off-diagonal equality follows from $-B_{ex}^*B_{ey}/\Delta=gC_{xy}$; the degree count gives the diagonal. Thus $-\sqrt{2}gD$ is not a higher-order nuisance that can be omitted. An explicitly additional compensation

$$V_c=+\sqrt{2}gD \tag{19}$$

cancels it to this order, leaving the irrelevant low-sector identity shift $-\sqrt{2}gI$. The compensation coefficient is derived from the balanced mediator couplings, not fitted to a desired final fidelity.

Density-assisted hopping with controllable Peierls phases has been realized using driven cold atoms [1]. This supports the physical operator class, not our particular 12 auxiliary states, compensation, numerical scales, or autonomous implementation. The construction here is a finite mediator candidate with local edge support, not a completed laboratory device or unique first-principles action.

5 High-energy residue changes the physical cost

Fix $\gamma = 0.03$, the same $g = 0.1$ and the same 25-update pulse. The three declared detunings test the mediator, not a fit of masses or pulse durations. Without compensation, the effective target fidelity is 0.95183145; with compensation it is 0.99960092. The full model gives

V_c present	Δ	full F_D	P_{aux}	E_{aux}	W_{net}
no	20	0.93561813	0.0249881	0.599715	0.496818
no	40	0.94438231	0.0125617	0.552714	0.499559
no	80	0.94935599	0.0042897	0.360335	0.340302
yes	20	0.98003504	0.0188301	0.451921	0.373277
yes	40	0.99051139	0.00886303	0.389973	0.351192
yes	80	0.99631112	0.00333585	0.280212	0.263537

Here $F_D = |(\langle D_+|, 0_{12})\Psi_f|^2$, not a fidelity renormalized after rejecting auxiliaries, and

$$E_{\mathrm{aux}} = (4 + \Delta)P_{\mathrm{aux}}, \qquad P_{\mathrm{sys}} + P_{\mathrm{aux}} = 1. \tag{20}$$

With $\mathcal{V} = \mathcal{H}_{\mathrm{on}} - \mathcal{H}_{\mathrm{off}}$, the complete switching ledger is

$$W_{\mathrm{on}} = \langle \Psi_i|\mathcal{V}|\Psi_i\rangle = \begin{cases} \sqrt{2}g/3 & \text{with compensation,} \\ 0 & \text{without,} \end{cases} \tag{21}$$

$$W_{\mathrm{off}} = -\langle \Psi_f|\mathcal{V}|\Psi_f\rangle, \tag{22}$$

$$W_{\mathrm{net}} = W_{\mathrm{on}} + W_{\mathrm{off}} = \langle \mathcal{H}_{\mathrm{off}}\rangle_f - \langle \mathcal{H}_{\mathrm{off}}\rangle_i. \tag{23}$$

The expectation of $\mathcal{H}_{\mathrm{on}}$ is conserved throughout the hold. Thus there is no missing energy: the controller supplies the positive net work displayed here. In the compensated $\Delta = 80$ case, only about 0.334% remains auxiliary, yet it carries energy 0.2802 (excess above the common energy 4 is $\Delta P_{\mathrm{aux}} = 0.2669$). The direct effective-pulse work -0.003331 cannot replace this ledger.

The origin is the sudden boundary. In the frame shifted by energy 4, writing the system and auxiliary amplitudes as $a(s), b(s)$ gives exactly

$$b(s) = -\mathrm{i}\int_0^s e^{-\mathrm{i}\Delta(s-u)} B a(u)\, du. \tag{24}$$

The dressed low-energy component and a rapidly oscillating boundary component are both present. Since $\|B\|$ scales as $\sqrt{g\Delta}$, an auxiliary probability of order g/Δ can carry energy of order g. Small population alone supplies no small-energy bound. This is not a claim of a monotonic or universal asymptotic coefficient.

Extending the existing readout by assigning every auxiliary branch to failure retains total probability one. Its success probability remains the unconditional F_D in the table. No bath, unreported sink or postselected energy accounting has been introduced.

6 Physical interpretation and the next decision

Three tasks must remain separate. Transport to a single outlet, coherent clock preparation and durable event recording have different criteria. G-R9b's chiral transfer reaches $|02\rangle$, a product state whose remote conditional density matrix $\rho_{B|\phi} = |2\rangle\langle 2|$ does not vary with A's phase reading. Its full cycle returns to the input after $3t_*$; direction is not irreversibility. The prepared D_+ instead supports G-R10's nontrivial conditional orbit. Neither construction alone generates an autonomous saved event.

The other scope correction concerns carriers: propagation does not require a gapless band. For example, $E(k) = \Delta_c + 2J(1 - \cos k)$ has $v_g = 2J \sin k/\hbar \neq 0$ despite $\Delta_c > 0$ (unit lattice spacing). This example does not derive a spatial network or a passing trigger; it only rejects a needless gaplessness prerequisite.

What is established. Equations (3)–(5) realize the preparation generator by local conditional hopping rather than a hand-inserted mode projector. The original contact is retained in all actual nine-state fixtures. Equations (15)–(23) give an explicit finite relay, its Stark compensation and an all-channel energy ledger. The high fidelity is a model prediction, not measured hardware performance. The inherited ring flux, new hopping quadrature and phase, balanced relay amplitudes, auxiliary detuning, pulse and controller remain physical inputs. Locality and symmetry permit them but do not select them uniquely. This is not an unconditional first-principles derivation.

A bounded next choice. The local-operator existence and finite-mediator audit can close here. For a controlled realization, choose one physical mediator/control implementation and test a single smooth on/off protocol, counting auxiliary energy and the compensation work together. For autonomous events, replace the imposed pulse by a specified passing trigger and include reflection, return and its energy. These are alternatives, not new mandatory papers before using the bounded result. No new mass fit, universal E_d law, time arrow or Route-F promotion follows.

Reproducibility. The new verifier has 189 passing checks: local support, exact block, symmetries, finite Peierls identity, original-contact dynamics, full work ledgers, Stark terms, and all auxiliary/readout branches. Matrix exponentiation and spectral propagation agree independently. The G-R9b and G-R11 code, cards and result hashes are pinned unchanged; their valid restricted statements are preserved, not rewritten as new claims. The source files are under Route G:

```
data/CORRELATED_PREPARATION_CARD.json
code/verify_gr12_correlated_preparation.py
output/gr12_correlated_preparation.json
```

References

[1] F. Görg et al., *Realization of density-dependent Peierls phases to engineer quantized gauge fields coupled to ultracold matter*, Nature Physics **15**, 1161–1167 (2019), arXiv:1812.05895, doi:10.1038/s41567-019-0615-4.